

Multilevel Security Based Access Control System

Id: V250E2 L1

Introduction

Security systems are essential for protecting sensitive areas such as lockers, laboratories, and restricted rooms. Traditional lock systems are vulnerable because they rely only on a physical key or single password authentication.

To improve security, this project implements a Multilevel Security Based Access Control System using LPC2138 ARM7 microcontroller. The system uses two levels of authentication:

- 1) Password verification through a keypad.
- 2) Password verification through a keypad

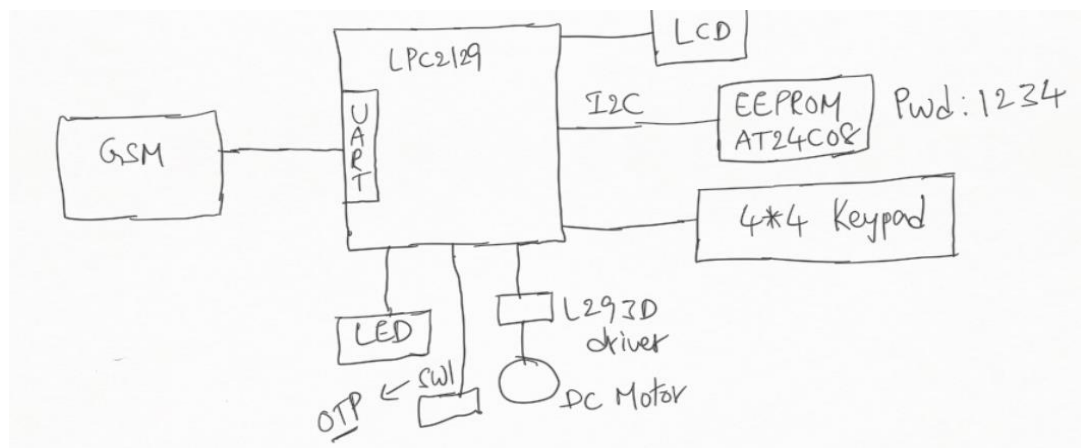
Objective

- The main objectives of this project are:
 - 1) To design a secure access control system
 - 2) To implement **two-factor authentication**
 - 3) To interface **keypad, LCD, UART, and motor driver** with LPC2148.
 - 4) To generate and verify **OTP for additional security**
 - 5) To display system status using **LCD display**

Hardware Components

- LPC2138 Microcontroller
- 4x4 Matrix Keypad
- 16x2 LCD Display
- DC Motor
- LED
- Push Button Switch
- Virtual Terminal
- Power supply

System Architecture



The working process of the system is as follows:

- 1) When the system starts, the LCD displays “**Multilevel Security Based Control System**”.
- 2) The system asks the user to **enter the password using the keypad**.
- 3) The entered password is compared with the stored password.
- 4) If the password is correct:
 - The user is prompted to **press the switch**.
 - The system generates a **random OTP**.
 - The OTP is sent to the **virtual terminal via UART**.
- 5) The user enters the OTP using the keypad.
- 6) The system compares the entered OTP with the generated OTP.
- 7) If the OTP is correct.
 - Access Granted.
 - LED turns ON
 - Motor runs to simulate door unlocking.
- 8) If incorrect password is entered **three times**, the system sends an **alert message**.

Codes:

main. C

```
#include <lcd.h>
#include <string.h>
#include "lcd.h"
#include "keypad.h"
#include "uart.h"
#include "delay.h"
#include "gsm.h"

#define LED    (1<<16)
#define MOTOR_IN1 (1<<17)
#define MOTOR_IN2 (1<<18)
#define SWITCH  (1<<11)

unsigned long seed=12345;

unsigned int generate_otp()
{
    seed = seed * 1103515245 + 12345;
    return (seed>>16)%9000 + 1000;
}

int main()
{
    char password[5]="7777";
    char entered[5];
    char otp_entered[5];
    char otp_string[5];
    unsigned int otp;
    int i;

    IO0DIR |= LED|MOTOR_IN1|MOTOR_IN2;
    IO0DIR &= ~SWITCH;
```

```

lcd_init();
uart_init();

lcd_print("Multilevel Security");
delay_ms(2000);

while(1)
{
    lcd_cmd(0x01);
    lcd_print("Enter Password");
    lcd_cmd(0xC0);

    for(i=0;i<4;i++)
    {
        char k=0;
        while(k==0) k=keypad_read();
        entered[i]=k;
        lcd_data(k);
    }
    entered[4]='\0';

    if(strcmp(entered,password)==0)
    {
        otp = generate_otp();

        otp_string[0]=(otp/1000)+'0';
        otp_string[1]=(otp/100)%10+'0';
        otp_string[2]=(otp/10)%10+'0';
        otp_string[3]=(otp%10)+'0';
        otp_string[4]='\0';

char sms_msg[30];

strcpy(sms_msg,"Your OTP: ");
strcat(sms_msg,otp_string);

gsm_send_sms(sms_msg);
        lcd_cmd(0x01);
        lcd_print("Enter OTP");
        lcd_cmd(0xC0);

        for(i=0;i<4;i++)
        {
            char k=0;
            while(k==0) k=keypad_read();
            otp_entered[i]=k;
            lcd_data(k);
        }
        otp_entered[4]='\0';

        if(strcmp(otp_entered,otp_string)==0)
        {
            lcd_cmd(0x01);
            lcd_print("Access Granted");

            IO0SET = MOTOR_IN1;
            IO0CLR = MOTOR_IN2;

            delay_ms(5000);

```

```

        IO0CLR = MOTOR_IN1;
    }
}
}
}

```

Uart. C

```

#include <lpc21xx.h>
#include "uart.h"

void uart_init()
{
    VPBDIV = 0x01;

    PINSEL0 |= 0x00000005;

    U0LCR = 0x83;
    U0DLL = 78;
    U0DLM = 0;
    U0LCR = 0x03;

    U0FCR = 0x07;
}

void uart_tx(char c)
{
    while(!(U0LSR & 0x20));
    U0THR = c;
}

void uart_string(char *s)
{
    while(*s)
        uart_tx(*s++);
}

```

Delay. h

```

#ifndef DELAY_H
#define DELAY_H

void delay_ms(unsigned int ms)
{
    unsigned int i,j;
    for(i=0;i<ms;i++)
        for(j=0;j<6000;j++);
}

#endif

```

LCD. h

```
#ifndef LCD_H
#define LCD_H

void lcd_init(void);
void lcd_cmd(unsigned char cmd);
void lcd_data(unsigned char data);
void lcd_print(char *s);

#endif
```

LCD. C

```
#include <lpc21xx.h>
#include "lcd.h"
#include "delay.h"

#define RS    (1<<8)
#define EN    (1<<9)
#define LCDDATA (0xFF<<16)

void lcd_cmd(unsigned char cmd)
{
    IO0CLR = RS;
    IO1CLR = LCDDATA;
    IO1SET = (cmd<<16);
    IO0SET = EN;
    delay_ms(2);
    IO0CLR = EN;
}

void lcd_data(unsigned char data)
{
    IO0SET = RS;
    IO1CLR = LCDDATA;
    IO1SET = (data<<16);
    IO0SET = EN;
    delay_ms(2);
    IO0CLR = EN;
}

void lcd_print(char *s)
{
    while(*s)
        lcd_data(*s++);
}
```

```

void lcd_init()
{
    IO0DIR |= RS | EN;
    IO1DIR |= LCDDATA;

    delay_ms(100);

    lcd_cmd(0x38);
    lcd_cmd(0x0C);
    lcd_cmd(0x06);
    lcd_cmd(0x01);
}

```

Keypad. H

```

#ifndef KEYPAD_H
#define KEYPAD_H

char keypad_read(void);

#endif

```

Keypad. C

```

#include <lpc21xx.h>
#include "keypad.h"
#include "delay.h"

/* Rows */
#define ROW1 (1<<27)
#define ROW2 (1<<28)
#define ROW3 (1<<29)
#define ROW4 (1<<30)

/* Columns */
#define COL1 (1<<12)
#define COL2 (1<<10)
#define COL3 (1<<14)
#define COL4 (1<<15)

#define ALL_ROWS (ROW1|ROW2|ROW3|ROW4)
#define ALL_COLS (COL1|COL2|COL3|COL4)

char keypad_read()
{
    PINSEL0 &= ~(0xFF<<20);

    IO0DIR |= ALL_ROWS;
    IO0DIR &= ~ALL_COLS;
}

```

```

IO0SET = ALL_ROWS;

/* Row1 */
IO0CLR = ROW1;
delay_ms(5);
if(!(IO0PIN & COL1)){while(!(IO0PIN & COL1));IO0SET=ROW1;return '7';}
if(!(IO0PIN & COL2)){while(!(IO0PIN & COL2));IO0SET=ROW1;return '8';}
if(!(IO0PIN & COL3)){while(!(IO0PIN & COL3));IO0SET=ROW1;return '9';}
IO0SET = ROW1;

/* Row2 */
IO0CLR = ROW2;
delay_ms(5);
if(!(IO0PIN & COL1)){while(!(IO0PIN & COL1));IO0SET=ROW2;return '4';}
if(!(IO0PIN & COL2)){while(!(IO0PIN & COL2));IO0SET=ROW2;return '5';}
if(!(IO0PIN & COL3)){while(!(IO0PIN & COL3));IO0SET=ROW2;return '6';}
IO0SET = ROW2;

/* Row3 */
IO0CLR = ROW3;
delay_ms(5);
if(!(IO0PIN & COL1)){while(!(IO0PIN & COL1));IO0SET=ROW3;return '1';}
if(!(IO0PIN & COL2)){while(!(IO0PIN & COL2));IO0SET=ROW3;return '2';}
if(!(IO0PIN & COL3)){while(!(IO0PIN & COL3));IO0SET=ROW3;return '3';}
IO0SET = ROW3;

/* Row4 */
IO0CLR = ROW4;
delay_ms(5);
if(!(IO0PIN & COL2)){while(!(IO0PIN & COL2));IO0SET=ROW4;return '0';}
IO0SET = ROW4;

return 0;
}

```

Uart.h

```

#ifndef UART_H
#define UART_H

void uart_init(void);
void uart_tx(char c);
void uart_string(char *s);

#endif

```

Uart. C

```
#include <lpc21xx.h>
#include "uart.h"

void uart_init()
{
    VPBDIV = 0x01;

    PINSEL0 |= 0x00000005;

    UOLCR = 0x83;
    UODLL = 78;
    UODLM = 0;
    UOLCR = 0x03;

    UOFCR = 0x07;
}

void uart_tx(char c)
{
    while(!(UOLSR & 0x20));
    UOTHR = c;
}

void uart_string(char *s)
{
    while(*s)
        uart_tx(*s++);
}
```

Gsm.h

```
#ifndef GSM_H
#define GSM_H

void gsm_send_sms(char *msg);
void send_sms_alert(void);

#endif
```

GSM.C

```
#include <lpc21xx.h>
```



```

#include "uart.h"
#include "gsm.h"

void gsm_send_sms(char *msg)
{
    uart_string("AT\r");
    delay_ms(1000);

    uart_string("AT+CMGF=1\r");
    delay_ms(1000);

    uart_string("AT+CMGS=\"+917995108589\"\r"); // change phone
number
    delay_ms(1000);

    uart_string(msg);

    uart_tx(26); // CTRL+Z
    delay_ms(3000);
}

void send_sms_alert()
{
    gsm_send_sms("Alert! 3 Wrong Password Attempts");
}

```

Result:

